
HPU BCA Semester V

E-COMMERCE

Unit-4: Securing the Business on Internet & CRM

Comprehensive Study Notes

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1. Security Policy, Procedures and Practices

1.1 Introduction to E-Commerce Security

Definition: E-commerce security refers to the measures taken to protect electronic transactions, customer data, and business information from unauthorized access, theft, damage, or disruption during online commercial activities.

Why Security is Critical for E-Commerce

Reason	Description
Customer Trust	Customers will not transact if they fear fraud
Financial Loss	Breaches lead to direct monetary loss
Reputation Damage	Security incidents destroy brand credibility
Legal Liability	Non-compliance leads to penalties and lawsuits
Business Continuity	Security ensures uninterrupted operations
Competitive Advantage	Strong security differentiates from competitors

1.2 Security Objectives (CIA Triad)

A. Confidentiality

Definition: Ensuring that information is accessible only to those authorized to access it.

Key Aspects: Data should be readable only by intended recipients; sensitive information must be protected from unauthorized viewing; applies to data in transit, at rest, and in use.

Technique	Description
Encryption	Converting data into unreadable format using algorithms
Access Controls	Restricting who can view certain information
Authentication	Verifying user identity before granting access
Data Masking	Hiding sensitive parts of data

B. Integrity

Definition: Ensuring that information is accurate, complete, and has not been altered or tampered with.

Key Aspects: Data cannot be modified without detection; transaction information must be accurate; messages must be authentic and unchanged.

Technique	Description
Hashing	Creating a unique digital fingerprint of data
Digital Signatures	Cryptographic proof of document authenticity
Checksums	Verification of data integrity

Audit Trails	Recording all changes to data
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C. Availability

Definition: Ensuring that information and services are accessible to authorized users when needed.

Key Aspects: 24x7 availability for e-commerce operations; quick recovery from failures; protection from denial-of-service attacks.

Technique	Description
Redundancy	Backup systems and servers
Disaster Recovery	Plans for system restoration
Load Balancing	Distributing traffic across servers
Firewalls	Protecting against attacks
Intrusion Detection	Monitoring for threats

1.3 Security Policy

Definition: A security policy is a formal document that outlines the rules, procedures, and guidelines for protecting an organization's information assets.

Components of a Security Policy

Section	Description
Policy Statement	High-level overview and objectives
Scope	Who and what the policy applies to
Definitions	Key terms and concepts
Responsibilities	Who is responsible for security
Access Control	How access is granted and revoked
Data Protection	Data classification and handling
Password Management	Password creation and maintenance
Network Security	Firewalls, VPN, wireless security
Physical Security	Building and hardware protection
Incident Response	What to do when breach occurs
Compliance	Regulatory requirements
Enforcement	Consequences of violations
Review	How often policy is updated

1.4 Security Procedures

Definition: Procedures are the step-by-step instructions that implement the security policy.

Types of Security Procedures

A. Administrative Procedures

- User account creation/deletion
- Access request and approval process
- Security awareness training
- Security incident reporting
- Background checks for employees

B. Technical Procedures

Procedure	Description
Backup Procedures	Regular data backup schedule and testing
Patch Management	Applying security updates to systems
Antivirus Updates	Regular virus definition updates
Log Review	Periodic review of system logs
Vulnerability Scanning	Regular security assessments
Password Change	Scheduled password updates

C. Physical Procedures

- Building access control
- Server room security
- Hardware disposal methods

- Visitor logging

1.5 Security Practices

A. Best Security Practices for E-Commerce

Practice	Description
PCI DSS Compliance	Meet Payment Card Industry Data Security Standards
Secure Sockets Layer (SSL)	Use HTTPS for all transactions
Two-Factor Authentication	Require additional verification beyond passwords
Regular Security Audits	Assess vulnerabilities periodically
Employee Training	Educate staff on security risks
Data Encryption	Encrypt sensitive data at rest and in transit
Firewall Configuration	Protect network perimeter
Intrusion Detection Systems	Monitor for suspicious activity
Incident Response Plan	Preparedness for security events
Third-Party Security	Vet vendors and partners

B. Common Security Mistakes to Avoid

Mistake	Why It's Dangerous	Solution
Weak Passwords	Easy to guess or crack	Strong password policies
Outdated Software	Known vulnerabilities exploited	Regular updates

No Encryption	Data intercepted during transmission	SSL/TLS encryption
Single Point of Failure	One breach compromises everything	Defense in depth
Ignoring Logs	Attacks go undetected	Regular log review
No Backup Strategy	Data loss leads to business failure	Automated backups
Too Much Trust	Insider threats exploited	Principle of least privilege

2. Transaction Security

2.1 Introduction to Transaction Security

Definition: Transaction security refers to the measures taken to ensure that electronic transactions are safe, confidential, and legally binding.

2.2 Authentication Methods

Definition: The process of verifying the identity of a user or entity.

Authentication Factors

Factor	Description	Examples
Knowledge	Something you know	Password, PIN
Possession	Something you have	OTP, Smart card
Inherence	Something you are	Biometrics

Authentication Types

Authentication Type	Examples	Security Level
Single-Factor	Password only	Low
Two-Factor (2FA)	Password + OTP	Medium
Multi-Factor (MFA)	Password + OTP + Biometric	High

Passwords and PINs

Characteristics of Strong Passwords:

- Minimum 8-12 characters

- Combination of upper/lowercase letters
- Include numbers and special characters
- Not based on personal information
- Changed regularly

Mistake	Example
Using common words	"password", "admin"
Using personal info	Birthdate, pet name
Same password everywhere	One breach compromises all
Writing down passwords	Sticky notes on monitor
Sharing passwords	Giving to colleagues
Using keyboard patterns	"qwerty123"

Biometric Authentication

Biometric Type	Description	Advantages	Disadvantages
Fingerprint	Unique fingerprint patterns	Fast, reliable	Can be spoofed
Face Recognition	Facial features	Contactless	Lighting affects accuracy
Iris Recognition	Eye patterns	Highly accurate	Expensive
Voice Recognition	Voice patterns	Natural, easy	Background noise affects
Palm/Finger Vein	Vascular patterns	Very secure	Specialized hardware

2.3 Authorization

Definition: The process of determining what resources an authenticated user is allowed to access.

Mechanism	Description	Example
Role-Based Access (RBAC)	Access based on user role	Admin, Manager, User
Rule-Based Access	Access based on rules	IP-based restrictions
Attribute-Based Access	Access based on attributes	Location, time, device
Mandatory Access	System-defined access	Military classification
Discretionary Access	Owner-defined access	File permissions

2.4 Encryption

Definition: The process of converting plain text into cipher text that is unreadable without the correct decryption key.

A. Symmetric Encryption

Aspect	Description
Definition	Same key used for encryption and decryption
Key Sharing	Both parties must share the key
Speed	Very fast
Key Management	Challenging (key distribution)

Examples	AES, DES, 3DES, Blowfish
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Advantages: Fast processing, good for large data volumes.

Disadvantages: Key distribution problem; everyone needs the same key.

B. Asymmetric Encryption (Public Key)

Aspect	Description
Definition	Different keys for encryption (public) and decryption (private)
Key Sharing	Public key can be shared widely
Speed	Slower than symmetric
Key Management	Easier (public keys distributed)
Examples	RSA, ECC, Diffie-Hellman

Advantages: Secure key exchange, digital signatures possible, no key sharing required.

Disadvantages: Slower processing, more computational overhead.

Digital Signatures

Definition: A digital signature is a mathematical scheme for verifying the authenticity and integrity of a digital message.

Feature	Description
Authenticity	Confirms identity of sender
Integrity	Confirms document not altered
Non-Repudiation	Sender cannot deny sending
Legal Validity	Recognized under law

Applications in E-Commerce: Contract signing, invoice approval, order confirmation, payment authorization.

2.5 Digital Certificates

Definition: A digital certificate is an electronic document that uses a digital signature to bind a public key with an identity.

Component	Description
Subject	Owner's name, organization, email
Public Key	Owner's public key
Issuer	Certificate Authority (CA)
Serial Number	Unique identifier
Validity Period	Start and end dates
Digital Signature	CA's signature
Certificate Version	X.509 version

Types of Digital Certificates

Type	Description	Use Case
SSL/TLS Certificate	Secures website communication	HTTPS websites
Code Signing Certificate	Verifies software authenticity	Application installation
Email Certificate	Secures email communication	S/MIME email
Client Certificate	Identifies individual users	Two-factor authentication
Device Certificate	Identifies devices	IoT authentication

2.6 SSL/TLS Protocol

Definition: Cryptographic protocols that provide secure communication over a computer network.

SSL/TLS in E-Commerce:

- Indicated by "https://" in URL
- Padlock icon in browser
- Encrypts all data between browser and server
- Essential for payment transactions

Benefits of SSL/TLS

Benefit	Description
Data Encryption	All data encrypted during transmission
Authentication	Server identity verified
Data Integrity	Data cannot be altered in transit
Trust Indicator	Visual cues build customer trust
SEO Benefit	Google prefers HTTPS sites

2.7 Secure Electronic Transaction (SET)

Definition: SET was a communications protocol standard for securing credit card transactions over networks, specifically the Internet.

Key Participants in SET

Participant	Role
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Cardholder	Customer making purchase
Merchant	Business selling product/service
Issuer	Cardholder's bank
Acquirer	Merchant's bank
Payment Gateway	Processes transaction

SET Features

Feature	Description
Confidentiality	Payment information encrypted
Integrity	Messages cannot be altered
Authentication	Cardholder and merchant authenticated
Non-Repudiation	Digital signatures used
Dual Signatures	Links order and payment info

SET vs SSL

Aspect	SET	SSL/TLS
Security Level	Very High	High
Implementation	Complex	Simple
Adoption	Limited	Widespread
Cost	Expensive	Affordable
Infrastructure	Requires special software	Built into browsers

3. Customer Relationship Management (CRM)

3.1 Introduction to CRM

Definition: Customer Relationship Management (CRM) is a business strategy that uses technology and processes to manage and analyze customer interactions and data throughout the customer lifecycle, with the goal of improving business relationships and customer retention.

Key Objectives of CRM

- Acquire new customers
- Retain existing customers
- Grow customer lifetime value
- Improve customer satisfaction
- Reduce marketing costs

3.2 Evolution of CRM

Era	Phase	Focus
1980s	Database Marketing	Customer data collection
1990s	Traditional CRM	Sales force automation
2000s	E-CRM	Web-based interaction
2010s	Social CRM	Social media integration
2020s	AI-Powered CRM	Predictive analytics, chatbots

3.3 E-CRM (Electronic CRM)

Definition: E-CRM is the application of internet and web technologies to customer relationship management. It uses digital channels to manage and improve customer relationships.

E-CRM vs Traditional CRM

Aspect	Traditional CRM	E-CRM
Channel	Phone, mail, in-person	Web, email, social media, mobile
Interaction Type	Company-initiated	Customer-initiated
Data Collection	Limited	Massive (Web analytics)
Response Time	Hours/Days	Real-time
Personalization	Limited	Highly personalized
Cost	High	Lower
Customer Reach	Limited	Global

E-CRM Key Features

Feature	Description
24x7 Availability	Customers can interact anytime
Self-Service	Customers help themselves online
Real-time Response	Immediate feedback
Personalization	Tailored experiences
Analytics	Deep customer insights

Automation	Automated processes
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3.4 Components of E-CRM

Component	Description	Functions
Operational CRM	Automates front-office business processes	Sales, marketing, service automation
Analytical CRM	Analyzes customer data for insights	Segmentation, predictive modeling, reporting
Collaborative CRM	Manages communication across channels	Multi-channel integration, interaction management

3.5 Applications of E-CRM

A. Marketing Automation

Application	Description	Tools
Email Campaigns	Automated email sequences	Mailchimp, Klaviyo
Lead Generation	Capture and nurture leads	Landing pages, forms
Segmentation	Group customers by attributes	Behavior, demographics
Personalization	Tailored content and offers	Recommendations
Social Media Marketing	Social engagement	Hootsuite, Buffer
A/B Testing	Test and optimize	Campaign optimization

B. Sales Automation

Application	Description	Tools
Contact Management	Store customer data	Salesforce, HubSpot
Lead Tracking	Monitor prospect progress	Pipeline management
Opportunity Management	Track potential sales	Deal tracking
Quote Generation	Create and send quotes	Automated quoting
Order Management	Process orders	E-commerce integration

C. Service Automation

Application	Description	Tools
Help Desk	Customer support tickets	Zendesk, Freshdesk
Live Chat	Real-time support	Chat widgets
Chatbots	AI-powered assistance	Virtual assistants
Knowledge Base	Self-service resources	Help center
Feedback Management	Collect and act on feedback	Surveys, NPS
Returns Management	Process returns	Returns portal

3.6 Benefits of E-CRM

Benefit	Description
Improved Customer Experience	Personalized, seamless interactions
Increased Customer Retention	Better understanding of customer needs
Higher Revenue	More effective marketing and sales
Cost Reduction	Automated processes, lower overhead
Better Decision Making	Data-driven insights
Competitive Advantage	Superior customer relationships
Customer Loyalty	Stronger emotional connection
Cross-Selling/Up-Selling	Identify sales opportunities

3.7 Challenges of E-CRM

Challenge	Description	Solution
Data Privacy	Collecting and using customer data	Compliance, transparency
Integration	Connecting different systems	API integration
Data Quality	Incomplete/inaccurate data	Data cleansing
User Adoption	Employees resist new systems	Training, change management
Cost	Implementation and maintenance	ROI justification

Security	Protecting customer data	Strong security measures
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4. The E-CRM Marketing in India

4.1 Indian E-CRM Market Overview

- CRM market in India growing rapidly
- Increasing adoption by SMEs
- Cloud-based CRM gaining popularity
- Mobile CRM adoption increasing
- AI and analytics integration becoming common

4.2 Major Trends in Indian E-CRM

Trend	Description	Examples
Digital-First Approach	Businesses prioritizing digital channels	E-commerce portals
Omni-Channel Engagement	Consistent experience across channels	Social media, email, chat
Personalization	Customized experiences	Recommendations, offers
Data Analytics	Leveraging customer data	Behavior analysis
AI Integration	Chatbots, predictive analytics	AI-powered suggestions
Cloud CRM	Scalable, cost-effective solutions	Salesforce, Zoho, HubSpot

4.3 Indian E-CRM Applications

Retail / E-Commerce

- Personalized product recommendations
- Abandoned cart recovery
- Customer loyalty programs
- Post-purchase engagement

Banking / Financial Services

- Personalized banking experiences
- Fraud detection
- Customer segmentation
- Targeted offers

Healthcare

- Patient relationship management
- Appointment scheduling
- Health record management
- Patient engagement

Education

- Student relationship management
- Course recommendations
- Alumni engagement
- Feedback collection

5. Global Scenario for E-CRM

5.1 Global E-CRM Market

Aspect	Detail
Market Size	Multi-billion dollar industry
Growth Rate	10-15% annually
Key Players	Salesforce, Oracle, Microsoft, SAP, HubSpot
Adoption Drivers	Digital transformation, customer expectations
Regional Leaders	North America, Europe, Asia-Pacific

5.2 Global E-CRM Trends

Trend	Description
AI and Machine Learning	Predictive analytics, chatbots
Social CRM	Integrating social media
Voice CRM	Voice interfaces
Mobile-First	Mobile-optimized CRM
Blockchain	Secure data management
Personalization at Scale	Hyper-personalization

5.3 Best Practices from Global Leaders

Practice	Description
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Customer-Centric Culture	Put customer at center of business
Data-Driven Decisions	Use analytics to guide strategy
Omni-Channel Experience	Seamless experience across channels
Continuous Innovation	Regularly update capabilities
Customer Feedback	Actively collect and act on feedback

6. CRM Utility in India

6.1 Why CRM is Important in India

Driver	Description
Large Customer Base	Over 1.4 billion population
Diverse Preferences	Varying needs across regions
Increasing Digital Adoption	More customers online
Competitive Market	Need to differentiate
Evolving Expectations	Rising customer demands

6.2 CRM Applications in Indian Context

A. E-Commerce (Flipkart, Amazon, Myntra)

Application	Example
Personalization	Product recommendations
Email Marketing	Abandoned cart emails
Customer Support	Chat, email, phone support
Loyalty Programs	Amazon Prime, Flipkart Plus
Customer Feedback	Reviews and ratings

B. Financial Services (Banks, Insurance)

Application	Example
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Onboarding	Account opening process
Transaction Management	UPI, mobile banking
Customer Support	Chatbots, call centers
Personalized Offers	Loan offers, investment tips
Feedback	NPS surveys

C. Telecommunications (Jio, Airtel, Vi)

Application	Example
Customer Onboarding	SIM activation
Service Management	Bill payments, recharges
Customer Support	24x7 support centers
Upselling	Plan recommendations
Churn Management	Retention offers

D. Travel and Hospitality (MakeMyTrip, Oyo)

Application	Example
Personalization	Personalized recommendations
Loyalty Programs	Points and rewards
Customer Support	Trip assistance
Feedback	Reviews and ratings

7. Security and CRM - Integrated View

7.1 Security Requirements for CRM

Requirement	Description
Data Privacy	Protect personal information
Access Control	Restrict who can view data
Encryption	Encrypt sensitive data
Audit Trails	Track data access
Compliance	Follow regulations (GDPR, DPDP)
Backup	Regular data backup
User Authentication	Secure login systems

7.2 Common Security Threats to CRM

Threat	Description	Mitigation
Data Breach	Unauthorized access to data	Encryption, access controls
Insider Threat	Employees misusing data	Monitoring, access restrictions
Phishing	Fake login pages	Training, awareness
Ransomware	Data held for ransom	Backup, security software
Cloud Vulnerabilities	Weak cloud security	Secure cloud configuration

7.3 Regulatory Compliance in India

Regulation	Description
DPDP Act (2023)	Digital Personal Data Protection Act
IT Act (2000)	Information Technology Act
IT (Reasonable Security Practices) Rules	Security requirements
PCI DSS	Payment card security
RBI Guidelines	Banking data security

8. Summary of Unit IV

Topic	Key Points
Security Policy	CIA Triad (Confidentiality, Integrity, Availability); Formal policy document
Security Procedures	Administrative, Technical, Physical
Security Practices	PCI DSS, SSL, 2FA, Encryption, Training
Authentication	Knowledge, Possession, Inherence; 2FA/MFA
Encryption	Symmetric (AES), Asymmetric (RSA); Digital signatures
Digital Certificates	SSL/TLS certificates; Certificate Authority
Transaction Security	SSL/TLS, SET
CRM	Customer Relationship Management
E-CRM	Online CRM using digital channels
E-CRM Components	Operational, Analytical, Collaborative
E-CRM Applications	Marketing, Sales, Service Automation
Global E-CRM	AI, Social CRM, Personalization
Indian E-CRM	E-commerce, Banking, Telecom, Travel
CRM Utility in India	Acquisition, Retention, Revenue, Cost Reduction

Key Security Threats Overview

Category	Threats
Malware	Virus, Worm, Trojan, Ransomware, Spyware, Adware
Hacking Attacks	Phishing, Spoofing, Session Hijacking, DoS/DDoS, SQL Injection, XSS, MITM

Security Technologies Overview

Technology	Purpose
Encryption	Symmetric (AES), Asymmetric (RSA), Hybrid
Firewalls	Packet Filter, Stateful, Proxy, NGFW
IDS/IPS	Intrusion Detection / Prevention
VPN	Virtual Private Network
2FA / Biometrics	Multi-factor authentication

9. Questions

Long Answer Questions (10 Marks)

Q1. Explain the security objectives (CIA Triad) in e-commerce.

Q2. What are security policies, procedures, and practices?
Explain with examples.

Q3. Discuss the various authentication methods used in e-commerce.

Q4. Explain encryption and digital signatures with diagrams.

Q5. What is CRM? Explain E-CRM and its applications.

Q6. Differentiate between traditional CRM and E-CRM.

Q7. Discuss the utility of CRM in Indian e-commerce.

Short Notes (5 Marks)

Q8. Write short notes on:

- Digital Certificates and PKI
- SSL/TLS Protocol
- SET Protocol
- Biometric Authentication

- Firewalls and IDS/IPS
- E-CRM Components
- Security Threats in E-Commerce
- Regulatory Compliance (DPDP Act, IT Act)
- E-CRM Marketing in India
- Global E-CRM Trends

Key Diagrams to Practice

1. CIA Triad - Confidentiality, Integrity, Availability
2. Security Policy Structure - Components flow chart
3. Encryption Process - Symmetric and Asymmetric
4. Digital Signature Process - Sender and Receiver flow
5. SSL/TLS Handshake - Client-Server interaction
6. E-CRM Components - Operational, Analytical, Collaborative
7. CRM Utility Diagram - Acquisition, Retention, Revenue

End of Unit IV - E-Commerce Study Notes (HPU BCA Sem V)